

Indian Institute of Technology Hyderabad

Signal Processing, Fall 2026

Quiz 3, 10.09.2026, 10 points

1. Give examples of *discrete-time signals* from the real-world that are *low-pass* and *high-pass* (one example each). Justify your answer. (2)
2. Give examples of *discrete-time systems* from the real-world that are *low-pass* and *high-pass* (one example each). Justify your answer. (2)
3. Recall from yesterday's class the definition of the DTFT: $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$, and the summation result $\sum_{k=M_1}^{M_2} \alpha^k = \frac{\alpha^{M_1} - \alpha^{M_2+1}}{1 - \alpha}$, $\alpha \neq 1$, $M_2 \geq M_1$. Use these expressions to show how the length of a 1-D *averaging filter* relates to frequency selectivity. (3)
4. Recall the 2-D averaging filter (of size $m \times m$, $m > 1$) from the image processing demo.
 - (a) Give one *non-zero* synthetic image example that the filter will allow to *pass through unchanged*. Justify your answer. (1.5)
 - (b) Give one *non-zero* synthetic image example for which the filter's *output is zero*. State any assumptions and justify your answer. (1.5)